Vignesh Annamalai

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Summary

Software Engineer with 4+ years of experience building resilient, high-availability distributed systems for large-scale enterprise environments. Engineered complex, user-facing post-transaction workflows at scale for Walmart, leveraging deep cross-stack expertise. Adept at full-lifecycle engineering, bridging robust UIs with enterprise Java/Spring ecosystems, authoring and profiling custom Go libraries.

Skills

Languages: Java, Golang, Python, JavaScript, TypeScript, C, C++, SQL, GraphQL, HTML/CSS
Frameworks: Spring Boot, Spring Cloud, Next.js, React.js, Node.js, mongoose
Databases, Caching & Messaging: MySQL, PostgreSQL, Cassandra, Redis, MongoDB, Kafka
DevOps & Tools: Docker, Jenkins, Toxiproxy, TestCafe, Jira, Git, CI/CD Pipelines
Architecture & Security: API Gateways, OAuth2 & JWT, Distributed Caching, Chaos/Resiliency Testing
Concepts: Geospatial Data Engineering, Spatial Indexing, Computer Vision, Agentic AI (MCP Servers & skills)

Experience

Software Developer Engineer 2 — July 2023 – Present

Walmart Global Tech — *Bengaluru, KA*

- **Engineered an on-demand Twilio masking architecture** for driver phone numbers within the Live Order Tracking flow. Designed a real-time validation layer that eliminated low-utilization pre-provisioning, reducing redundant API calls by 90% and generating **$400K/year** in operational cost savings.
- **Architected a dynamic Customer Choice UI** for the Live Shop flow in Walmart.com, aggregating four distinct APIs (personalization, search, typeahead, and amends) into a seamless order substitution workflow.
- **Modernized the Go ecosystem** by refactoring the go-spymemcached library to support externalized configurations via CCM, enabling zero-downtime deployments without manual code changes.
- **Fortified system reliability** by implementing Toxiproxy for network resiliency testing, simulating latency, timeouts, and connection drops to validate application fault tolerance under adverse conditions.
- **Developed an Astro API-powered MCP Server** with integrated AI skills to fully automate the creation of complex test-order states, broadcasting results directly to Slack, and saving hundreds of hours in manual QA setup.
- **Built and scaled a custom Jira REST API extension** adopted by engineering and product managers across the organization to automate Epic/Story creation, sizing, and standardized labeling.

Software Engineer Internship — January 2023 – June 2023

Walmart Global Tech — *Bengaluru, KA*

- **Reduced upstream backend traffic by ∼49%** (3.1M to 1.58M daily calls) by engineering a dynamic polling configuration for Live Order Tracking and optimizing GraphQL payload sizes by half.
- **Championed Web Tech Debt & Release Quality** by engineering a TestCafe-based browser automation script integrated with Jenkins-based cron jobs for zero-touch validation of all P1 flows during bi-weekly release sign-offs.

Projects

Radius (Location-Based Community Feed) — April 2026 – Present

- **Microservice Architecture:** Architected and developed a scalable, location-based community feed application, separating client-facing routing from core business logic using Spring Boot and Next.js.
- Engineered a Spring Cloud API Gateway for centralized OAuth2 authentication, eliminating XSS vulnerabilities by converting HttpOnly frontend sessions into stateless backend JWTs.
- Integrated a distributed Redis cache (via Spring Session) with the API Gateway to decouple session memory from compute instances, ensuring zero-downtime deployments and persistent user sessions across container restarts.
- Built a responsive, dynamic user interface utilizing Next.js Server-Side Rendering (SSR) for enhanced SEO and robust asset caching for optimal performance.
- **Integrated PostgreSQL with the PostGIS extension**, leveraging Geography data types and GiST indexing to perform highly optimized, scalable ST_DWithin radius queries, eliminating full-table scans for location-based feeds.
- **Tech Stack:** *Java, Spring Boot, Spring Cloud, Next.js, Redis, PostgreSQL, Docker*

Automated CCTV Data Logging & Recognition System **July 2022 – November 2022**

- **Engineered a computer vision pipeline** utilizing a custom Haar cascade for subject detection and a novel Conditional Generative Adversarial Network (cGAN) to restore and enhance pixelated, low-light Regions of Interest (ROI) for accurate recognition.
- Architected an automated data ingestion backend that processed raw CCTV footage, accurately identifying subjects and persisting timestamped instances into a cloud-hosted, distributed MongoDB database.
- Developed a full-stack web application to query the distributed database and dynamically generate chronological tracking timelines for specific subjects, effectively eliminating hours of manual video foraging and streamlining security reviews.
- **Tech Stack:** *Python, OpenCV, TensorFlow, PyTorch, Node.js, React, MongoDB*

Awards & Recognitions

- **Bravo Award, Walmart (2025):** Awarded for driving large-scale performance optimizations, architecting resilient system enhancements, and championing comprehensive technical debt resolution across the post-transaction domain.
- **3x Walmart Excellence Badges (2023 – 2024):** Awarded for driving GraphQL performance optimizations, cross-functional team collaboration, and building developer productivity tools.
- **Runner-Up, Caterpillar CodeAthon (2022):** Placed 2nd among 563 teams by developing an automated spatial tracking solution for commercial environments; leveraged OpenCV and Point-in-Polygon algorithm to monitor real-time fleet movement across construction site perimeters.

Education

Bachelor of Technology, Information Technology **2019 – 2023**

Anna University, Madras Institute of Technology (MIT), Chennai, TN *CGPA: 9.19 / 10*